

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12405435
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Huegerich; Thomas P. et al.

Power and optical fiber interface

Abstract

A power and optical fiber interface system includes a housing having an interior. A cable inlet is configured to receive a hybrid cable having an electrical conductor and an optical fiber. An insulation displacement connector (IDC) is situated in the interior of the housing configured to electrically terminate the conductor, and a cable outlet is configured to receive an output cable that is connectable to the IDC and configured to output signals received via the optical fiber.

Inventors: Huegerich; Thomas P. (Fuquay Varina, NC), Kachmar; Wayne M. (North Bennington, VT), Flaig; Robert Charles (Lancaster, PA), Bretz; Dwight A. (Hummelstown, PA)

Applicant: COMMSCOPE TECHNOLOGIES LLC (Hickory, NC)

Family ID: 59088348

Assignee: CommScope Technologies LLC (Claremont, NC)

Appl. No.: 18/750567

Filed: June 21, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20250060547 A1	Feb. 20, 2025

Related U.S. Application Data

continuation parent-doc US 18200154 20230522 ABANDONED child-doc US 18750567
continuation parent-doc US 17567331 20220103 US 11656418 20230523 child-doc US 18200154
continuation parent-doc US 16704964 20191205 US 11215776 20220104 child-doc US 17567331
continuation parent-doc US 15985068 20180521 US 10502912 20191210 child-doc US 16704964
continuation parent-doc US 15373709 20161209 US 9977208 20180522 child-doc US 15985068
continuation parent-doc US 14331873 20140715 US 9557505 20170131 child-doc US 15373709
continuation-in-part parent-doc WO PCT/US2014/030969 20140318 PENDING child-doc US 14331873
us-provisional-application US 61846392 20130715
us-provisional-application US 61802989 20130318

Publication Classification

Int. Cl.: G02B6/44 (20060101); G02B6/42 (20060101); H04L12/10 (20060101); H05K5/02 (20060101); G02B6/38 (20060101); H01B11/22 (20060101)

U.S. Cl.:

CPC **G02B6/4416** (20130101); **G02B6/426** (20130101); **G02B6/4278** (20130101); **G02B6/4284** (20130101);
G02B6/4448 (20130101); **H04L12/10** (20130101); **H05K5/0204** (20130101); G02B6/3817 (20130101);
G02B6/4431 (20230501); H01B11/22 (20130101)

Field of Classification Search

CPC: G02B (6/4416); H01B (11/22)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4089585	12/1977	Slaughter et al.	N/A	N/A
4199225	12/1979	Slaughter et al.	N/A	N/A
4220812	12/1979	Ney et al.	N/A	N/A
4359598	12/1981	Dey et al.	N/A	N/A
4365865	12/1981	Stiles	N/A	N/A
4401361	12/1982	Slaughter	N/A	N/A
4420220	12/1982	Dean et al.	N/A	N/A
4467138	12/1983	Brorein	N/A	N/A
4497537	12/1984	Dench	N/A	N/A
4552432	12/1984	Anderson et al.	N/A	N/A
4595839	12/1985	Braun et al.	N/A	N/A
4695127	12/1986	Ohlhaber et al.	N/A	N/A
4723832	12/1987	Okazato et al.	N/A	N/A
4729628	12/1987	Kraft et al.	N/A	N/A
4761053	12/1987	Cogelia et al.	N/A	N/A
4787705	12/1987	Shinmoto et al.	N/A	N/A
4844575	12/1988	Kinard et al.	N/A	N/A
4852965	12/1988	Mullin et al.	N/A	N/A
4867527	12/1988	Dotti et al.	N/A	N/A
4895426	12/1989	Pinson	N/A	N/A
5109457	12/1991	Panuska et al.	N/A	N/A
5138685	12/1991	Arroyo et al.	N/A	N/A
5155304	12/1991	Gossett et al.	N/A	N/A
5268971	12/1992	Nilsson et al.	N/A	N/A
5369518	12/1993	Aslami et al.	N/A	N/A
5448670	12/1994	Blew et al.	N/A	N/A
5469523	12/1994	Blew et al.	N/A	N/A
5494461	12/1995	Bippus et al.	N/A	N/A
5539851	12/1995	Taylor et al.	N/A	N/A
5555336	12/1995	Winslow	N/A	N/A
5555338	12/1995	Haag et al.	N/A	N/A
5557698	12/1995	Gareis et al.	N/A	N/A
5636308	12/1996	Personne et al.	N/A	N/A
5651081	12/1996	Blew et al.	N/A	N/A
5677974	12/1996	Elms et al.	N/A	N/A
5737470	12/1997	Nagano et al.	N/A	N/A
5745627	12/1997	Arroyo et al.	N/A	N/A
5778116	12/1997	Tomich	N/A	N/A
5778652	12/1997	Kunze	N/A	N/A
5838858	12/1997	White	N/A	N/A
5896480	12/1998	Scharf et al.	N/A	N/A
5913003	12/1998	Arroyo et al.	N/A	N/A
5982966	12/1998	Bonicel	N/A	N/A
6088499	12/1999	Newton et al.	N/A	N/A
6101305	12/1999	Wagman et al.	N/A	N/A
6110213	12/1999	Vinciarelli et al.	N/A	N/A
6142802	12/1999	Berg et al.	N/A	N/A

6169834	12/2000	Keller	N/A	N/A
6195487	12/2000	Anderson et al.	N/A	N/A
6236789	12/2000	Fitz	N/A	N/A
6343172	12/2001	Schiestle	N/A	N/A
6347172	12/2001	Keller et al.	N/A	N/A
6363192	12/2001	Spooner	N/A	N/A
6370303	12/2001	Fitz et al.	N/A	N/A
6463198	12/2001	Coleman	N/A	N/A
6493491	12/2001	Shen et al.	N/A	N/A
6501888	12/2001	Gimblet et al.	N/A	N/A
6542674	12/2002	Gimblet	N/A	N/A
6563990	12/2002	Hurley et al.	N/A	N/A
6567592	12/2002	Gimblet et al.	N/A	N/A
6599025	12/2002	Deutsch	N/A	N/A
6621964	12/2002	Quinn et al.	N/A	N/A
6687437	12/2003	Starnes et al.	N/A	N/A
6714710	12/2003	Gimblet	N/A	N/A
6719461	12/2003	Cull	N/A	N/A
6738547	12/2003	Spooner	N/A	N/A
6744954	12/2003	Tanaka et al.	N/A	N/A
6755575	12/2003	Kronlund et al.	N/A	N/A
6758693	12/2003	Inagaki et al.	N/A	N/A
6792184	12/2003	Conrad et al.	N/A	N/A
6836603	12/2003	Bocanegra et al.	N/A	N/A
6845200	12/2004	Quinn	N/A	N/A
6847767	12/2004	Hurley et al.	N/A	N/A
6931183	12/2004	Panak et al.	N/A	N/A
6965718	12/2004	Koertel	N/A	N/A
7006740	12/2005	Parris	N/A	N/A
7085459	12/2005	Conrad et al.	N/A	N/A
7158703	12/2006	Mjelstad	N/A	N/A
7184634	12/2006	Hurley et al.	N/A	N/A
7218821	12/2006	Bocanegra et al.	N/A	N/A
7225534	12/2006	Kachmar	N/A	N/A
7272281	12/2006	Stahulak et al.	N/A	N/A
7310430	12/2006	Mallya et al.	N/A	N/A
7362936	12/2007	Stark et al.	N/A	N/A
7371014	12/2007	Willis et al.	N/A	N/A
7387517	12/2007	Kusuda et al.	N/A	N/A
7401985	12/2007	Aronson et al.	N/A	N/A
7445389	12/2007	Aronson	N/A	N/A
7454107	12/2007	Miller et al.	N/A	N/A
7467896	12/2007	Melton et al.	N/A	N/A
7494287	12/2008	Wang et al.	N/A	N/A
7499616	12/2008	Aronson et al.	N/A	N/A
7532796	12/2008	Chiasson	N/A	N/A
7539380	12/2008	Abernathy et al.	N/A	N/A
7567741	12/2008	Abernathy et al.	N/A	N/A
7643631	12/2009	Kennedy	N/A	N/A
7643713	12/2009	Büthe et al.	N/A	N/A
7692098	12/2009	Wyatt, II et al.	N/A	N/A
7693375	12/2009	Freeland et al.	N/A	N/A
7712976	12/2009	Aronson et al.	N/A	N/A
7744288	12/2009	Lu et al.	N/A	N/A
7762727	12/2009	Aronson	N/A	N/A
7778510	12/2009	Aronson et al.	N/A	N/A
7796853	12/2009	Abernathy et al.	N/A	N/A
7837397	12/2009	Fingler et al.	N/A	N/A
7876989	12/2010	Aronson et al.	N/A	N/A
7897873	12/2010	Gemme et al.	N/A	N/A

7918609	12/2010	Melton et al.	N/A	N/A
8041166	12/2010	Kachmar	N/A	N/A
8059929	12/2010	Cody et al.	N/A	N/A
8083417	12/2010	Aronson et al.	N/A	N/A
8155525	12/2011	Cox	N/A	N/A
8175433	12/2011	Caldwell et al.	N/A	N/A
8184935	12/2011	Kachmar	N/A	N/A
8204348	12/2011	Keller et al.	N/A	N/A
8224141	12/2011	Kachmar et al.	N/A	N/A
8238706	12/2011	Kachmar	N/A	N/A
8244087	12/2011	Sales Casals et al.	N/A	N/A
8249410	12/2011	Andrus et al.	N/A	N/A
8270838	12/2011	Cox	N/A	N/A
8290320	12/2011	Kachmar	N/A	N/A
8297854	12/2011	Bickham et al.	N/A	N/A
8301003	12/2011	de los Santos Campos et al.	N/A	N/A
8328433	12/2011	Furuyama	N/A	N/A
8358893	12/2012	Sanderson et al.	N/A	N/A
8363994	12/2012	Kachmar	N/A	N/A
8376630	12/2012	Wang et al.	N/A	N/A
8480312	12/2012	Smith et al.	N/A	N/A
8538216	12/2012	Abernathy et al.	N/A	N/A
8682124	12/2013	Logan	N/A	N/A
8712200	12/2013	Abernathy et al.	N/A	N/A
8761564	12/2013	Su et al.	N/A	N/A
9274302	12/2015	Gimblet et al.	N/A	N/A
9472314	12/2015	Kachmar et al.	N/A	N/A
9557505	12/2016	Huegerich et al.	N/A	N/A
9837186	12/2016	Kachmar et al.	N/A	N/A
9893811	12/2017	Chappell et al.	N/A	N/A
9977208	12/2017	Huegerich et al.	N/A	N/A
10014958	12/2017	Sipes, Jr. et al.	N/A	N/A
10171180	12/2018	Sipes, Jr. et al.	N/A	N/A
10247893	12/2018	Elkayam et al.	N/A	N/A
10277329	12/2018	Sipes, Jr. et al.	N/A	N/A
10277330	12/2018	Sipes, Jr.	N/A	N/A
10502912	12/2018	Huegerich et al.	N/A	N/A
10855381	12/2019	Sipes, Jr.	N/A	N/A
11025345	12/2020	Sipes, Jr. et al.	N/A	N/A
11215776	12/2021	Huegerich et al.	N/A	N/A
11656418	12/2022	Huegerich et al.	N/A	N/A
2002/0015565	12/2001	Imamura	N/A	N/A
2002/0126967	12/2001	Panak et al.	N/A	N/A
2002/0136510	12/2001	Heinz et al.	N/A	N/A
2002/0147978	12/2001	Dolgonos et al.	N/A	N/A
2003/0108351	12/2002	Feinberg et al.	N/A	N/A
2003/0202756	12/2002	Hurley et al.	N/A	N/A
2003/0215197	12/2002	Simon et al.	N/A	N/A
2004/0258165	12/2003	Peltonen	N/A	N/A
2005/0002622	12/2004	Sutehall et al.	N/A	N/A
2005/0185903	12/2004	Koertel	N/A	N/A
2006/0045443	12/2005	Blazer	N/A	N/A
2006/0153516	12/2005	Napiorkowski et al.	N/A	N/A
2006/0165355	12/2005	Greenwood et al.	N/A	N/A
2006/0239183	12/2005	Robitaille et al.	N/A	N/A
2006/0246772	12/2005	Yamaguchi et al.	N/A	N/A
2006/0291787	12/2005	Seddon	N/A	N/A
2007/0248358	12/2006	Sauer	N/A	N/A
2007/0269170	12/2006	Easton et al.	N/A	N/A
2008/0037941	12/2007	Mallya et al.	N/A	N/A

2008/0131132	12/2007	Solheid et al.	N/A	N/A
2008/0219621	12/2007	Aldeghi et al.	N/A	N/A
2008/0260389	12/2007	Zheng	N/A	N/A
2008/0290729	12/2007	Schoenberg et al.	N/A	N/A
2008/0310848	12/2007	Yasuda et al.	N/A	N/A
2009/0003792	12/2008	Thompson	N/A	N/A
2009/0041413	12/2008	Hurley	N/A	N/A
2009/0097855	12/2008	Thelen et al.	N/A	N/A
2009/0297104	12/2008	Kachmar	N/A	N/A
2010/0007220	12/2009	Lo	N/A	N/A
2010/0080563	12/2009	DiFonzo et al.	N/A	N/A
2010/0150556	12/2009	Soto et al.	N/A	N/A
2010/0200270	12/2009	Gemme et al.	N/A	N/A
2010/0209058	12/2009	Ott	N/A	N/A
2010/0290787	12/2009	Cox	N/A	N/A
2010/0321591	12/2009	Onomatsu	N/A	N/A
2011/0021069	12/2010	Hu et al.	N/A	N/A
2011/0038582	12/2010	DiFonzo et al.	N/A	N/A
2011/0091174	12/2010	Kachmar	N/A	N/A
2011/0150401	12/2010	Furuyama	N/A	N/A
2011/0188815	12/2010	Blackwell, Jr. et al.	N/A	N/A
2011/0217010	12/2010	Kachmar	N/A	N/A
2011/0268446	12/2010	Cune et al.	N/A	N/A
2011/0268452	12/2010	Beamon et al.	N/A	N/A
2011/0280527	12/2010	Tamura	N/A	N/A
2011/0293227	12/2010	Wu	N/A	N/A
2011/0311191	12/2010	Hayashishita et al.	N/A	N/A
2012/0008904	12/2011	Han et al.	N/A	N/A
2012/0008905	12/2011	Han et al.	N/A	N/A
2012/0008906	12/2011	Han et al.	N/A	N/A
2012/0057821	12/2011	Aronson et al.	N/A	N/A
2012/0080225	12/2011	Kim et al.	N/A	N/A
2012/0114288	12/2011	Wu	N/A	N/A
2012/0141064	12/2011	Walker	N/A	N/A
2012/0191997	12/2011	Miller	N/A	N/A
2012/0230637	12/2011	Kachmar	N/A	N/A
2012/0242168	12/2011	Tsai et al.	N/A	N/A
2012/0281952	12/2011	McColloch	N/A	N/A
2012/0281953	12/2011	Choi et al.	N/A	N/A
2012/0288245	12/2011	Hurley et al.	N/A	N/A
2012/0295486	12/2011	Petersen et al.	N/A	N/A
2012/0315004	12/2011	Register, III et al.	N/A	N/A
2013/0011106	12/2012	Congdon, II et al.	N/A	N/A
2013/0022318	12/2012	Fingler et al.	N/A	N/A
2013/0058613	12/2012	Jost et al.	N/A	N/A
2013/0094821	12/2012	Logan	N/A	N/A
2013/0094823	12/2012	Gimblet et al.	N/A	N/A
2013/0173939	12/2012	Peto	N/A	N/A
2013/0287349	12/2012	Faulkner et al.	N/A	N/A
2013/0330086	12/2012	Berlin et al.	N/A	N/A
2014/0016527	12/2013	Coffey	N/A	N/A
2014/0072264	12/2013	Schroder et al.	N/A	N/A
2014/0087742	12/2013	Brower et al.	N/A	N/A
2014/0258742	12/2013	Chien et al.	N/A	N/A
2014/0314412	12/2013	Soto et al.	N/A	N/A
2014/0338968	12/2013	Kachmar	N/A	N/A
2015/0078740	12/2014	Sipes, Jr. et al.	N/A	N/A
2015/0125146	12/2014	Erreygers et al.	N/A	N/A
2015/0304742	12/2014	Siddhaye et al.	N/A	N/A
2015/0309271	12/2014	Huegerich et al.	N/A	N/A

2015/0378118	12/2014	Huegerich et al.	N/A	N/A
2016/0020911	12/2015	Sipes, Jr. et al.	N/A	N/A
2016/0050029	12/2015	Huegerich et al.	N/A	N/A
2016/0294475	12/2015	Chappell et al.	N/A	N/A
2016/0300643	12/2015	Kachmar et al.	N/A	N/A
2017/0184799	12/2016	Huegerich et al.	N/A	N/A
2018/0061529	12/2017	Kachmar et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2008309134	12/2008	AU	N/A
2010201456	12/2009	AU	N/A
2010235866	12/2009	AU	N/A
101449189	12/2008	CN	N/A
201359882	12/2008	CN	N/A
101925841	12/2009	CN	N/A
201868149	12/2010	CN	N/A
102136649	12/2010	CN	N/A
102263340	12/2010	CN	N/A
102934377	12/2012	CN	N/A
103428056	12/2012	CN	N/A
0 629 889	12/1993	EP	N/A
1 650 888	12/2005	EP	N/A
2 393 220	12/2010	EP	N/A
2 393 222	12/2010	EP	N/A
2 676 161	12/2012	EP	N/A
2468274	12/2012	GB	N/A
10-145299	12/1997	JP	N/A
10-1997-0060748	12/1996	KR	N/A
1999-0040389	12/1998	KR	N/A
20-0273482	12/2001	KR	N/A
2010/102201	12/2009	WO	N/A
2011/146720	12/2010	WO	N/A
2012/058181	12/2011	WO	N/A
2012/112532	12/2011	WO	N/A
2013/117598	12/2012	WO	N/A

OTHER PUBLICATIONS

European Search Report for Application No. 14808346.2 mailed Oct. 26, 2016. cited by applicant

European Search Report for Application No. 14797917.3 mailed Dec. 6, 2016. cited by applicant

Extended European Search Report for Application No. 14825675.3 mailed Jan. 30, 2017. cited by applicant

Extended European Search Report for Application No. 14751600.9 mailed Mar. 22, 2017. cited by applicant

Extended European Search Report for Application No. 21168354.5 mailed Aug. 23, 2021. cited by applicant

“GE—Critical Power, DAS and Small Cell Powering Architectures,” DAS and Small Cell Congress, May 1, 2013; 22 pages. cited by applicant

IEC 60603-7-7, Second Edition Jun. 2006, “Connectors for electronic equipment”, Part 7-7. cited by applicant

International Search Report and Written Opinion for PCT/US2014/015969 mailed Jul. 21, 2014. cited by applicant

International Search Report and Written Opinion for PCT/US2014/000115 mailed Sep. 24, 2014. cited by applicant

International Search Report and Written Opinion for PCT/US2014/046678 mailed Nov. 6, 2014. cited by applicant

International Search Report and Written Opinion for PCT/US2014/030969 mailed Nov. 26, 2014. cited by applicant

Littelfuse Inc., “Transient Suppression Devices and Principles” application note AN9768, dated Jan. 1998. Retrieved from [http://www.littelfuse.com/~media/](http://www.littelfuse.com/~media/electronics_technical/application_notes/varistors/littelfuse_transient_suppression_devices_and_principles_application_note.pdf)

[electronics_technical/application_notes/varistors/littelfuse_transient_suppression_devices_and_principles_application_note.pdf](http://www.littelfuse.com/~media/electronics_technical/application_notes/varistors/littelfuse_transient_suppression_devices_and_principles_application_note.pdf) cited by applicant

OneReach PoE Extender System brochure from Berk-Tek, 8 pages (Jan. 2011). cited by applicant

Silvertel, “Ag5700 200W Powered Device Module” datasheet, dated Aug. 2012. Retrieved from

<http://www.silvertel.com/images/datasheets/Ag5700-datasheet-200W-Power-over-Ethernet-PoE-PD.pdf>. cited by applicant

European Examination Report for EP Application No. 21168354.5 mailed Oct. 17, 2024. cited by applicant

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application of U.S. Non-Provisional patent application Ser. No. 18/200,154, filed May 22, 2023, which is a continuation application of U.S. patent application Ser. No. 17/567,331, filed Jan. 3, 2022, now U.S. Pat. No. 11,656,418, issued May 23, 2023, which is a continuation application of U.S. patent application Ser. No. 16/704,964, filed Dec. 5, 2019, now U.S. Pat. No. 11,215,776, issued Jan. 2, 2022, which is a continuation application of U.S. Non-Provisional patent application Ser. No. 15/985,068, filed May 21, 2018, now U.S. Pat. No. 10,502,912, issued Dec. 10, 2019, which is a continuation of U.S. Non-Provisional patent application Ser. No. 15/373,709, filed Dec. 9, 2016, now U.S. Pat. No. 9,977,208, issued May 22, 2018, which is a continuation application of U.S. Non-Provisional patent application Ser. No. 14/331,873, filed Jul. 15, 2014, now U.S. Pat. No. 9,557,505, issued Jan. 31, 2017, which claims the benefit of U.S. Provisional Patent Application Ser. No. 61/846,392, filed Jul. 15, 2013 and is a continuation-in-part of PCT Patent Application No. PCT/US2014/030969, filed Mar. 18, 2014, which claims the benefit of U.S. Provisional Patent Application Ser. No. 61/802,989, filed Mar. 18, 2013. The disclosures of all of the above-mentioned patent applications are hereby incorporated by reference in their entireties.

BACKGROUND

(1) The present disclosure relates generally to hybrid optical fiber and electrical communication systems.
(2) Rapid growth of portable high-speed wireless transceiver devices (e.g., smart phones, tablets, laptop computers, etc.) continues in today's market, thereby creating higher demand for untethered contact. Thus, there is growing demand for integrated voice, data and video capable of being transmitted wirelessly at data rates of 10 Gbits/second and faster. To provide the bandwidth needed to support this demand will require the cost effective and efficient deployment of additional fixed location transceivers (i.e., cell sites or nodes) for generating both large and small wireless coverage areas. Fiber optic technology is becoming more prevalent as service providers strive to deliver higher bandwidth communication capabilities to customers/subscribers. The phrase “fiber to the x” (FTTX) generically refers to any network architecture that uses optical fiber in place of copper within a local distribution area. Example FTTX networks include fiber-to-the-node (FTTN) networks, fiber-to-the-curb (FTTC) networks, fiber-to-the-home (FTTH), and more generally, fiber-to-the-wireless (FTTW).

SUMMARY

(3) In accordance with aspects of the present disclosure, examples of a power and optical fiber interface system include a housing having an interior. A cable inlet is configured to receive a hybrid cable having an electrical conductor and an optical fiber. An insulation displacement connector (IDC) is situated in the interior of the housing configured to electrically terminate the conductor, and a cable outlet is configured to receive an output cable that is connectable to the IDC and configured to output signals received via the optical fiber.
(4) In accordance with further aspects of the disclosure, examples of the disclosed system include a power converter, such as a DC-DC converter electrically connected to the IDC. An optical fiber management device, such as an optical splice device, is situated in the interior of the housing and configured to receive the optical fiber. A media board is included in some embodiments that is configured to convert optical signals to electrical signals. In some implementations, the IDC includes a housing with a fiber pass-through groove configured to route optical fibers through the housing of the IDC, and first and second conductor grooves are situated on either side of the fiber pass-through groove to receive first and second conductors.
(5) Another aspect of the present disclosure relates to a powered fiber optic system. The powered fiber optic system includes a first location including a power source and fiber optic network access and a plurality of active devices remotely positioned with respect to the first location. The powered fiber optic system further includes a plurality of hybrid cables routed from the first location toward the active devices. The hybrid cables include optical fibers for transmitting optical signals and electrical conductors for carrying power. The powered fiber optic system further includes interface devices mounted adjacent to the active devices for providing interfaces between the hybrid cables and the active devices. The interface devices include electrical power management circuitry positioned within the closure for providing DC-to-DC voltage conversion within the closure and also include circuit protection circuitry for providing current surge protection.
(6) A further aspect of the present disclosure relates to an interface device for providing an interface between a hybrid cable and an active device. The interface device includes a closure adapted for outside environmental use and a cable anchoring structure for securing a hybrid cable to the closure. The hybrid cable is configured to carry both electrical power and optical signals. The interface device also includes electrical power management circuitry positioned within the closure for providing DC-to-DC voltage conversion within the closure. The electrical power management circuitry is customizable to output one of a plurality of different DC voltage levels such that the DC output level can be matched with a power requirement of the active device. The interface device also includes electrical protection circuitry

positioned within the closure and an output configuration for outputting power and communications signals from the interface device to the active device. The output configuration has a format that is customizable and selectable from a plurality of formats that include all of the following formats: a) a power over Ethernet format or a power over Ethernet plus format; and b) a format including one or more optical fibers for the optical signals and separate electrical conductors for power.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a system diagram showing an example distribution of wireless coverage areas deployed using a power and optical fiber interface system in accordance with principles of the present disclosure.
- (2) FIG. 2 is a transverse cross-sectional view of a power/optical fiber hybrid cable in accordance with principles of the present disclosure.
- (3) FIG. 3 is a perspective view of a portion of the hybrid cable of FIG. 2 with electrically conductive portions of the cable showing separated from a central optical fiber portion of the cable.
- (4) FIG. 4 is a plan view of the hybrid cable of FIGS. 2 and 3 with the electrically conductive portions of the hybrid cable trimmed relative to the central fiber optic portion of the hybrid cable.
- (5) FIG. 5 is a transverse cross-sectional view of another power/optical fiber hybrid cable in accordance with principles of the present disclosure.
- (6) FIG. 6 is a block diagram conceptually illustrating aspects of a communication and power distribution system in accordance with principles of the present disclosure.
- (7) FIG. 7 is a top view of an interface device in accordance with principles of the present disclosure.
- (8) FIG. 8 is a perspective view of the interface device shown in FIG. 7.
- (9) FIG. 9 is a partial top view of the interface device shown in FIG. 7, illustrating aspects of an insulation displacement connector (IDC) in an open position.
- (10) FIG. 10 is another partial top view of the interface device shown in FIG. 7, illustrating aspects of the IDC in a closed position.
- (11) FIG. 11 is a perspective view of the interface device shown in FIG. 7, illustrating an embodiment that includes an optical splice device.
- (12) FIG. 12 is a top view of the interface device shown in FIG. 11.
- (13) FIG. 13 is a top view of the interface device shown in FIG. 7, illustrating an embodiment that includes a media board.
- (14) FIG. 14 is a circuit diagram illustrating an example power conditioning circuit.
- (15) FIG. 15 shows a system in accordance with the principles of the present disclosure having a rack mounted power supply.
- (16) FIG. 16 shows a hybrid cable system in accordance with the principles of the present disclosure.

DETAILED DESCRIPTION

(17) In the following Detailed Description, reference is made to the accompanying drawings, which form a part hereof, and in which is shown by way of illustration specific embodiments in which the invention may be practiced. In this regard, directional terminology, such as top, bottom, front, back, etc., is used with reference to the orientation of the Figure(s) being described. Because components of embodiments can be positioned in a number of different orientations, the directional terminology is used for purposes of illustration and is in no way limiting. It is to be understood that other embodiments may be utilized and structural or logical changes may be made without departing from the scope of the present invention. The following detailed description, therefore, is not to be taken in a limiting sense.

(18) FIG. 1 shows a system 10 in accordance with the principles of the present disclosure for enhancing the coverage areas provided by cellular technologies (e.g., GSM, CDMA, UMTS, LTE, WiMax, WiFi, etc.). The system 10 includes a base location 11 (i.e., a hub) and a plurality of wireless coverage area defining equipment 12a, 12b, 12c, 12d, 12e and 12f (sometimes collectively referred to as equipment 12 herein) distributed about the base location 11. In certain examples, the base location 11 can include a structure 14 (e.g., a closet, hut, building, housing, enclosure, cabinet, etc.) protecting telecommunications equipment such as racks, fiber optic adapter panels, passive optical splitters, wavelength division multiplexers, fiber splice locations, optical fiber patching and/or fiber interconnect structures and other active and/or passive equipment. In the depicted example, the base location 11 is connected to a central office 16 or other remote location by a fiber optic cable such as a multi-fiber optical trunk cable 18 that provides high band-width two-way optical communication between the base location 11 and the central office 16 or other remote location. In the depicted example, the base location 11 is connected to the wireless coverage area defining equipment 12a, 12b, 12c, 12d, 12e and 12f by hybrid cables 20. The hybrid cables 20 are each capable of transmitting both power and communications between the base location 11 and the wireless coverage area defining equipment 12a, 12b, 12c, 12d, 12e and 12f.

(19) The wireless coverage area defining equipment 12a, 12b, 12c, 12d, 12e and 12f can each include one or more wireless transceivers 22. The transceivers 22 can include single transceivers 22 or distributed arrays of transceivers 22.

As used herein, a “wireless transceiver” is a device or arrangement of devices capable of transmitting and receiving wireless signals. A wireless transceiver typically includes an antenna for enhancing receiving and transmitting the wireless signals. Wireless coverage areas are defined around each of the wireless coverage area defining equipment **12a**, **12b**, **12c**, **12d**, **12e** and **12f**. Wireless coverage areas can also be referred to as cells, cellular coverage areas, wireless coverage zones, or like terms. Examples of and/or alternative terms for wireless transceivers include radio-heads, wireless routers, cell sites, wireless nodes, etc.

(20) In the depicted example of FIG. 1, the base location **11** is shown as a base transceiver station (BTS) located adjacent to a radio tower **24** supporting and elevating a plurality the wireless coverage area defining equipment **12a**. In one example, the equipment **12a** can define wireless coverage areas such as a macrocells or microcells (i.e., cells each having a coverage area less than or equal to about 2 kilometers wide). The wireless coverage area defining equipment **12b** is shown deployed at a suburban environment (e.g., on a light pole in a residential neighborhood) and the equipment **12c** is shown deployed at a roadside area (e.g., on a roadside power pole). The equipment **12c** could also be installed at other locations such as tunnels, canyons, coastal areas, etc. In one example, the equipment **12b**, **12c** can define wireless coverage areas such as microcells or picocells (i.e., cells each having a coverage area equal to or less than about 200 meters wide). The equipment **12d** is shown deployed at a campus location (e.g., a university or corporate campus), the equipment **12e** is shown deployed at a large public venue location (e.g., a stadium), and the equipment **12f** is shown installed at an in-building or near-building environment (e.g., multi-dwelling unit, high rise, school, etc.). In one example, the equipment **12d**, **12e**, and **12f** can define wireless coverage areas such as microcells, picocells, or femtocells (i.e., cells each having a coverage area equal to or less than about 10 meters wide).

(21) The wireless coverage area defining equipment **12** are often located in areas without power outlets conveniently located. As noted above, the hybrid cable **20** provides both power and data to the equipment **12**. FIG. 2 is a transverse cross-sectional view taken through an example of one of the hybrid cables **20** of FIG. 1. Hybrid cable **20** includes an outer jacket **200** having a transverse cross-sectional profile that defines a major axis **202** and a minor axis **204**. The outer jacket has a height H measured along the minor axis **204** and a width W measured along the major axis **202**. The width W is greater than the height H such that the transverse cross-sectional profile of the outer jacket **200** is elongated along the major axis **202**.

(22) The outer jacket **200** can include a left portion **206**, a right portion **208** and a central portion **210**. The left portion **206**, the right portion **208** and the central portion **210** can be positioned along the major axis **202** with the central portion **210** being disposed between the left portion **206** and the right portion **208**. The left portion **206** can define a left passage **212**, the right portion **208** can define a right passage **214** and the central portion **210** can define a central passage **216**. The passages **212**, **214** and **216** can have lengths that extend along a central longitudinal axis **218** of the cable **20** for the length of the cable. A left electrical conductor **220** is shown positioned within the left passage **212**, a right electrical conductor **222** is shown positioned within the right passage **214** and at least one optical fiber **224** is shown positioned within the central passage **216**. Certain embodiments include from 1 to 12 fibers **224**, for example. The left electrical conductor **220**, the right electrical conductor **222** and the optical fiber **224** have lengths that extend along the central longitudinal axis **218** of the cable **20**.

(23) Still referring to FIG. 2, the hybrid cable **20** includes a left pre-defined tear location **226** positioned between the central portion **210** and the left portion **206** of the outer jacket **200**, and a right pre-defined tear location **228** positioned between the central portion **210** and the right portion **208** of the outer jacket **200**. The left pre-defined tear location **226** is weakened such that the left portion **206** of the outer jacket **200** can be manually torn from the central portion **210** of the outer jacket **200**. Similarly, the right pre-defined tear location **228** is weakened such that the right portion **208** of the outer jacket **200** can be manually torn from the central portion **210** of the outer jacket **200**. The left pre-defined tear location **226** is configured such that the left portion **206** of the outer jacket **200** fully surrounds the left passage **212** and the central portion **210** of the outer jacket **200** fully surrounds the central passage **216** after the left portion **206** of the outer jacket **200** has been torn from the central portion **210** of the outer jacket **200**. In this way, the left electrical conductor **220** remains fully insulated and the optical fiber **220** remains fully protected after the left portion **206** has been torn from the central portion **210**. The right pre-defined tear location **228** is configured such that the right portion **208** of the outer jacket **200** fully surrounds the right passage **214** and the central portion **210** of the outer jacket **200** fully surrounds the central passage **219** after the right portion **208** of the outer jacket **200** has been torn from the central portion **210** of the outer jacket **200**. In this way, the right electrical conductor **222** remains fully insulated and the optical fiber **224** remains fully protected after the right portion **208** has been torn from the central portion **210**.

(24) FIG. 3 shows the hybrid cable **20** with both the left portion **206** and the right portion **208** torn away from the central portion **210**. In this configuration, both the left electrical conductor **220** and the right electrical conductor **222** are fully insulated by their corresponding left and right portions **206**, **208**. Additionally, the central portion **210** has a rectangular transverse cross-sectional shape that fully surrounds the central passage **216** so as to protect the optical fiber or fibers **224**.

(25) It will be appreciated that the left and right electrical conductors **220**, **222** have a construction suitable for carrying electricity. It will be appreciated that the electrical conductors can have a solid or stranded construction. Example sizes of the electrical conductors include 12 gauge, 16 gauge, or other sizes.

(26) The outer jacket **200** is preferably constructed of a polymeric material. In one example, the hybrid cable **20** and the

outer jacket **200** are plenum rated. In certain examples, the outer jacket **200** can be manufactured of a fire-retardant plastic material. In certain examples, the outer jacket **200** can be manufactured of a low smoke zero halogen material. Example materials for the outer jacket include polyvinyl chloride (PVC), fluorinated ethylene polymer (FEP), polyolefin formulations including, for example, polyethylene, and other materials.

(27) The central passage **216** can contain one or more optical fibers **224**. In certain examples, the optical fibers **224** can be coated optical fibers having cores less than 12 microns in diameter, cladding layers less than 240 microns in diameter, and coating layers less than 300 microns in diameter. It will be appreciated that the core and cladding layers typically include a silica based material. In certain examples, the cladding layer can have an index of a refraction that is less than the index of refraction of the core to allow optical signals that are transmitted through the optical fibers to be confined generally to the core. It will be appreciated that in certain examples, multiple cladding layers can be provided. In certain examples, optical fibers can include bend insensitive optical fibers having multiple cladding layers separated by trench layers. In certain examples, protective coatings (e.g., a polymeric material such as actelate) can form coating layers around the cladding layers. In certain examples, the coating layers can have diameters less than 300 microns, or less than 260 microns, or in the range of 240 to 260 microns. In certain examples, the optical fibers **224** can be unbuffered. In other examples, the optical fibers can include a tight buffer layer, a loose buffer layer, or a semi-tight buffer layer. In certain examples, the buffer layers can have an outer diameter of about 800 to 1,000 microns. The optical fibers can include single mode optical fibers, multi-mode optical fibers, bend insensitive fibers or other fibers. In still other embodiments, the optical fibers **224** can be ribbonized.

(28) As shown at FIG. 4, the left and right portions **206**, **208** can be trimmed relative to the central portion **210** after the left and right portions **206**, **208** have been torn away from the central portion **210**. In this configuration, the central portion **210** extends distally beyond the ends of the left and right portions **206**, **208**. In certain examples, insulation displacement connectors can be used to pierce through the jacket materials of the left and right portions **206**, **208** to electrically connect the left and right electrical connectors **220**, **222** to an electrical power source, ground, active components or other structures. It will be appreciated that the optical fibers **224** can be connected to other fibers with mechanical or fusion splices, or directly terminated with optical connectors. In other examples, connectorized pigtails can be spliced to the ends of the optical fibers **224**.

(29) Referring back to FIG. 2, the outer jacket **200** includes a top side **230** and a bottom side **232** separated by the height H. As depicted, the top and bottom sides **230**, **232** are generally parallel to one another. Each of the left and right pre-defined tear locations **226**, **228** includes an upper slit **234** that extends downwardly from the top side **230**, a lower slit **236** that extends upwardly from the bottom side **232** and a non-slitted portion **238** positioned between the upper and lower slits **234**, **236**. In one example embodiment, the upper and lower slits **234**, **236** are partially re-closed slits. In the depicted embodiment, the left and right pre-defined tear locations **226**, **228** also include jacket weakening members **240** that are imbedded in the non-slitted portions **238**. By way of example, the jacket weakening members **240** can include strands, monofilaments, threads, filaments or other members. In certain examples, the jacket weakening members **240** extend along the central longitudinal axis **218** of the cable **20** for the length of the cable **20**. In certain examples, the jacket weakening members **240** are aligned along the major axis **202**. In certain examples, the upper and lower slits **230**, **236** as well as the jacket weakening member **240** of the left pre-defined tear location **226** are aligned along a left tearing plane PL that is oriented generally perpendicular relative to the major axis **202**. Similarly, the upper and lower slits **234**, **236** as well as the jacket weakening member **240** of the right pre-defined tear location **228** are aligned along a right tearing plane PR that is oriented generally perpendicular with respect to the major axis **202**.

(30) Referring again to FIG. 2, the hybrid cable **20** can include a tensile strength structure **242** that provides tensile enforcement to the hybrid cable **20** so as to prevent tensile loads from being applied to the optical fibers **224**. In certain embodiments, the tensile strength structure **242** can include reinforcing structures such as Aramid yarns or other reinforcing fibers. In still other embodiments, the tensile strength structure **242** can have an oriented polymeric construction. In still other examples, a tensile strength structure **242** can include a reinforcing tape. In certain examples, the reinforcing tape can be bonded to the outer jacket **200** so as to line the central passage **216**. In certain examples, no central buffer tube is provided between the optical fibers **224** and the tensile reinforcing structure **242**. In certain examples, the tensile strength structure **242** can include a reinforcing tape that extends along the length of the hybrid cable **20** and has longitudinal edges/ends **244** that are separated so as to define a gap **244** therein between. In use, the tensile strength member **242** can be anchored to a structure such as a fiber optic connector, housing or other structure so as to limit the transfer of tensile load to the optical fibers **224**. It will be appreciated that the tensile strength structure **242** can be anchored by techniques such as crimping, adhesives, fasteners, bands or other structures.

(31) FIG. 5 shows an alternative hybrid cable **20'** having the same construction as the hybrid cable **20** except two tensile strength structures **242A**, **242B** have been provided within the central passage **216**. Tensile strength members **242A**, **242B** each include a tensile reinforcing tape that is bonded to the central portion **210** of the outer jacket **200**. The tensile strength members **242A**, **242B** can include portions that circumferentially overlap one another within the central passage **216**. In certain examples, by stripping away an end portion of the central portion **210**, the tensile strength structures **242A**, **242B** can be exposed and readily secured to a structure such as a fiber optic connector, a panel, a housing or other structure. In one example, the tensile strength structures **242A**, **242B** can be crimped, adhesively secured or otherwise attached to rods (e.g., epoxy rods reinforced with fibers) that are in turn secured within a

ruggedized fiber optic connector such as the fiber optic connector disclosed at U.S. Pat. No. 7,744,288 which is hereby incorporated by reference in its entirety, or the fiber optic connector disclosed at U.S. Pat. No. 7,918,609, which is hereby incorporated by reference in its entirety.

(32) As noted above, the electrical conductors **220, 222** could be 12 gauge (AWG) or 16 gauge, for example. In certain examples, a 12 gauge conductor **220, 220** provides up to 1175 meter reach at 15 W, and a 750 meter reach for 25 W devices. The 16 gauge implementations can provide reduced cost for shorter reach applications or lower power devices, for example.

(33) Providing power to remote active devices such as the wireless coverage area defining equipment **12** is often difficult and expensive. Providing required power protection and backup power further complicates powering such remote devices. Optical Network Terminals (ONT's) and Small Cell devices (such as picocells and metrocells) have “similar” power requirements. For example, 25 W, 12 VDC or 48 VDC devices are common, although variations occur. FIG. **6** conceptually illustrates an example of a communication signal and power distribution system **300** in accordance with aspects of the present disclosure. Among other things, the system **300** provides a simple, “universal” connection of the optical fiber **224** and electrical conductors **220,222** of the hybrid cable **20** to the equipment **12**.

(34) The system **300** includes a fiber patch panel **302** that terminates optical fibers carrying signals to be distributed to the desired wireless coverage area defining equipment **12** via the optical fibers **224** of the hybrid cables **20**. A power supply **304** connects to the conductors **220, 222** of the desired hybrid cable **20**. In some examples, the power supply **304** receives 120/220 VAC and provides 48 VDC nominal. In some embodiments, the fiber patch panel **302** and power supply **304** are rack mounted.

(35) A first end **306** of the hybrid cable **20** is connected to the appropriate optical fibers from the fiber patch panel **302** and to the power supply **304**. A second, distant end **308** of the cable **20** is connected to an interface device **310**. The interface device is connected to the wireless equipment **12**, either directly or through a media converter **312**. Examples of the interface **310** provide simplified termination of the hybrid cable **20**, allowing factory or field installation. In some embodiments, a DC-DC converter provides the desired voltage level for the particular device **12** to which it is connected and compensates for IR loss across variable link lengths.

(36) FIGS. **7-13** illustrate various views of embodiments of the interface device **310**. The interface device **310** includes a body **320** having an exterior **322** and an interior **324**. A cover **326** is connected to the body **320** by a hinge **328** such that the interface device **310** can be opened to expose the interior **324** for access by an operator. A mounting bracket **340** extends from the body **320** for mounting the interface device **310** as desired using screws or bolts, for example. In one example, the interface device **310** defines footprint dimensions of about 55 mm×125 mm×190 mm.

(37) A cable clamp **342** cooperates with the body **320** to fix the hybrid input cable **20** and an output cable **344** to the interface device **310** at a cable inlet **350** and a cable outlet **352**, respectively. As noted above, the hybrid cable **20** includes electrical conductors **220, 220** for supplying power to the interface device **310**, and ultimately the remote device **12**. In the illustrated examples, the interface device **310** includes an insulation displacement connector (IDC) **360** situated in the interior **324** of the interface device body **320** for connecting the conductors **220, 220** to the interface device **310**. Generally, an IDC (also sometimes referred to as insulation displacement termination and insulation-piercing connector) is an electrical connector that connects to one or more conductors of an insulated conductor by a connection process that forces a selectively sharpened blade or blades through the insulation to contact the conductor, eliminating the need to strip the insulation before connecting. Further, the connector blades cold weld to the conductors to form a gas-tight connection.

(38) FIGS. **9** and **10** illustrate the IDC **360** in closed and open positions, respectively. As shown in FIG. **3** and discussed in conjunction therewith above, the hybrid cable **20** is configured such that the left portion **206** and the right portion **208** can be torn away from the central portion **210**. In this configuration, both the left electrical conductor **220** and the right electrical conductor **222** are fully insulated by their corresponding left and right portions **206, 208**. Referring to FIG. **10**, the IDC **360** includes a housing **358** with first and second conductor grooves **362, 364** positioned on either side of a fiber pass-through groove **366**. Correspondingly, the electrical conductors **220, 222** of the hybrid cable **20** are situated on either side of the central portion **210** containing the optical fibers **224**.

(39) The conductors **220, 222** are received by the corresponding conductor grooves **362, 364**, and insulator clamping ribs **368** are situated to press against the jacket **200** to hold the hybrid cable **20** in place. The IDC **360** includes a cover **370** hingedly connected to the housing **358** that when closed presses IDC terminals **372** against the conductors **220, 222** and through the left portion **206** and the right portion **208** of the outer jacket **200** to make an electrical connection with the conductors **220, 222**. The illustrated IDC terminals **372** are angled to provide a gas-tight connection. In the illustrated example, the left and right portions **206, 208** are trimmed such that the conductors **220,222** extend beyond the IDC terminals **372** but remain within the housing **358** of the IDC **360**.

(40) In other embodiments, the IDC is configured in a “pass-through” power arrangement, wherein the terminals **372** pierce the left and right portions **206, 208** to contact the conductors **220,222**, but the left and right portions **206, 208** are not trimmed so they extend through the IDC **360** to be routed to equipment **12** or another interface device **310**, such as via the cable outlet **352**.

(41) In the illustrated example, a power converter **376**, such as a DC-DC voltage converter, is situated in the interior **324** of the base **320** and electrically connected to the IDC **360** so as to electrically connect the conductors **220, 220** of the

hybrid cable **20** to the power converter **376** via the IDC **360**. Thus, power entering the interface device **310** via the hybrid cable **20** can be conditioned and/or converted to the desired level for the wireless coverage area defining equipment **12** to which the interface device **310** connects. The power converter **376** is connectable to the output cable **344** to route the conditioned/converted power from the interface device **310** to the desired wireless equipment **12**. For instance, conductors of the output cable **344** could connect directly to the power converter **376** using screw terminals **378** thereon. In alternative embodiments, the power converter **376** can be omitted or bypassed if the power received by the interface device **310** is appropriate for the particular end device **12**. Further power connection arrangements are discussed herein below.

(42) The optical fibers **224** from the hybrid cable **20** are received by the centrally positioned fiber pass-through groove **366** to route the optical fibers **224** through the housing **358** of the IDC **360**. The fibers **224** extend from the housing **358** and are routed along the perimeter of the interior **324** of the interface device **310**. In some embodiments, the optical fibers **324** are routed through the interior **324** directly to the cable outlet **352**, along with a separate power output cable. More typically, the fibers **324** would be routed to a fiber management device **380**. In the illustrated example, fiber guides **374** are situated in the corners of the interior **324** for routing the optical fibers **324** in the interface device while maintaining a desired bend radius. In certain implementations, the optical fibers **324** are thus received at the cable inlet **350**, routed through the IDC housing **358** and the interior **324** of the interface device body **320** to the fiber management device **380**.

(43) FIGS. **11** and **12** illustrate an example of the interface device **310** wherein the fiber management device **380** includes an optical splice device for making a mechanical or fusion splice, for example. The illustrated fiber management device **380** thus includes furcation tubes **382** situated on splice holders **384**. In other implementations, other fiber optic management devices such as fiber optic connectors are provided. A strength member termination **386** is further provided in the embodiment illustrated in FIGS. **11** and **12**. The optical fibers **224** can thus be spliced, for example, to a fiber optic pig tail and routed to the cable outlet **352**. In some examples, the output cable **244** is also a hybrid cable including optical fibers that are spliced to the fibers **224** using the fiber management device **380**, and conductors that receive power from the power converter **376**.

(44) FIG. **13** illustrates another embodiment where the fiber management device **380** includes a media board **390** that converts optical signals received via the optical fibers **224** to electrical signals. A fiber optic connector **392**, such as an LC duplex input connector, is connected to the media board **390** to terminate the optical fibers routed through the interior **324** of the interface device **310** and receive optical signals therefrom. The media board **390** is electrically connected to the IDC **360**, either directly, or as in the illustrated embodiment, via the power converter **376**. In this manner, the media board is powered by power from the conductors **220,222** terminated by the IDC **360**. Additionally, in some embodiments, the media board **390** connects output power and electrical communication signals to a power over Ethernet (POE) connection. In such embodiments, the output cable **344** is a standard RJ-45 data/power cable that connects to a PoE jack **394** on the media board **390**. The RJ-45 cable can then be connected to the desired wireless coverage area defining equipment **12** to provide both communication signals and power thereto.

(45) Some embodiments, for example, include 12 fibers **224** situated in the central passage **216**. Typically, two optical fibers **224** are terminated in a given interface device **310**. Since the two fibers **224** carrying signals for the desired wireless equipment **12** are to be terminated in the interface device **310**, they are cut downstream of the interface device **310**. A slit can be cut in the central portion **216** providing an opening through which the desired fibers **224** can be pulled from the central portion and routed to the fiber management device **380**. The remaining optical fibers **224** remain intact within the central portion **216**, and can be passed through the interface device **310** to another device, for example.

(46) The power converter **276** provides DC/DC conversion, for example, as well as other power management functions such as circuit overload protection, mains cross protection, lightning protection, etc. In one particular embodiment, a 30 W, 12V output DC-DC converter from CUI Inc. of Tualatin, OR (P/N VYC30 W-Q48-S12-T) is used. Other DC-DC converters may be employed based on electrical requirements, packaging, etc. In some implementations, a conditioning circuit is integrated into the interface **310** to minimize voltage ripple. FIG. **14** shows an example of a typical conditioning circuit **400**. In further examples, overvoltage protection, such as a gas-tube, is incorporated between the IDC terminals and DC-DC converter input.

(47) Referring back to FIG. **6**, the fiber patch panel **302** can receive can receive optical signals from a remote location via a fiber optic trunk cable. Optical fibers of the trunk cable **110** can be separated at a fan-out device, or optical power splitters or wavelength division multiplexers can be used to split optical communications signals from the trunk cable to multiple optical fibers. The fibers can be routed to the patch panel **302**, and then to a desired one of the hybrid cables **20**, along with electrical power from the power supply **304**. In one example, the power supply **304** receives 120 volt or 220 volt alternating current. In one example, power supply **302** includes an AC/DC converter that converts the electrical power from alternating current to direct current. The power supply **304** converts the electrical power from the first voltage (e.g., 120v or 220v) to a second voltage that is less than the first voltage. In one example, the second voltage is less than or equal to 60 volts and 100 Watts such that the output voltage complies with NEC Class II requirements.

(48) The hybrid cable **20** can be used to transmit electrical power and optical communication signals from the fiber patch panel **302** and power supply **304** located at a first to the wireless equipment **12** located at a second location. The first end **306** of the hybrid cable **20** can include a first interface for connecting the hybrid cable to electrical power and

fiber optic communication at a connector, and the second end **308** of the hybrid cable **20** is received at the cable inlet **350** of the interface device **310**. The power converter **376** of the interface device **310** converts electrical power carried by the hybrid cable **20**, for example, to a direct current third voltage that is less than the second voltage. In one example, the third voltage corresponds to an electrical voltage requirement of the device **12**. In one example, the third voltage is 12V, 24V or 48V.

(49) In some implementations, a converter **312** is associated with the equipment **12** for converting optical signals to electrical signals. In such implementations, the optical fibers and power are provided from the interface device **310** to the converter **312**, which provides power and communication signals to the equipment **12**. In other implementations, the interface device **310** converts the optical signals to electrical signals using the media board **390**, and provides power and electrical communication signals to the equipment **12**.

(50) Aspects of the present disclosure relate to powered fiber cable systems capable of simultaneously powering and communicating with wireless coverage area defining equipment (e.g., transceivers, wireless routers, WiFi access points/WiFi hot spots, small cell devices, or like devices). The powered fiber cable system can also be used to power and communicate with other devices such as digital signage, high definition surveillance cameras, and like devices. Moreover, powered fiber cable systems in accordance with the principles of the present disclosure can be incorporated into fiber optic networks (e.g., fiber-to-the-home (FTTH), fiber-to-the-premises (FTTP, fiber-to-the-anything (FTTX)) to provide back-up power or primary power to optical network terminals (ONT) including electronics for providing optical-to-electrical conversion at or near a subscriber location. By providing back-up power using a powered fiber cable system in accordance with the principles of the present disclosure, battery back-ups at the optical network terminals can be eliminated. Powered fiber cable systems in accordance with the principles of the present disclosure are particularly well suited for supporting active devices at outdoor locations where power is not readily available. However, powered fiber cable systems in accordance of the principles of the present disclosure can also be used to support indoor applications such as in local area networks where power and fiber are provided to desk-top locations (e.g., fiber-to-the-desk (FTTD)). Other applications for powered fiber cable systems in accordance with the principles of the present disclosure relate to power-over-Ethernet extensions (PoE or PoE+).

(51) Aspects of the present disclosure relate to systems that provide a “rack to device” vision for both powering and communicating with active devices such as small cell devices, ONT's, WiFi hot spots, digital signage, surveillance cameras or like devices in one cable system. FIG. **15** shows an example powered fiber cable system **400** in accordance with the principles of the present disclosure. The powered fiber cable system **400** includes a rack **402** positioned at a location where power (e.g., a power system\grid that typically provides AC power such as a mains power system) and fiber network are available. Such locations where power and fiber optic network communications are available can be referred to as head ends. In certain examples, the power locations (i.e., head ends) can be co-located at a cell site base station, a base station on a building top, in a telecom closet or data center or anywhere where power and access to a fiber optic network are available. As shown at FIG. **15**, a patch panel **404** is mounted on the rack **402**. The patch panel **404** is coupled to a fiber optic network. For example, optical fibers optically corresponding to fiber optic distribution or feed cables can be connectorized and plugged into fiber optic adapters supported at the patch panel **404**. The rack **402** is also shown supporting a power supply unit **406** that in certain examples provides a DC output of 48 volts or less.

(52) In one example, the power supply can include a power express class II power converter shelf manufactured by General Electric. The power supply can include an AC/DC transformer for transforming alternating current (AC) from a mains power supply into DC power for distribution to remotely located active devices. In certain examples, the power supply **406** can power up to 32 hybrid cables **20** in a modular design with four modules and eight cables per module. In certain examples, the power supply is configured to output relatively low voltage DC current (e.g., less than or equal to 48 volts DC). In certain examples, the power source is National Electric Code (NEC) class 2 (as specified by Article 725) and Safety Extra Low Voltage (SELV) compliant. In certain examples, the voltage between any two conductors should not exceed 60 volts DC under normal operating conditions. In certain examples, the power supply is limited to 100 VA. Such low voltage circuits are advantageous because electricians are not required to install such systems, such systems are inherently safe because of the low voltage limits, and such systems can be installed in a conduit-free manner. In certain examples, the power supply can also include circuit protection electronics such as gas discharge tubes, metal oxide varistor components and transient voltage suppression structures/diodes.

(53) As shown at FIG. **15**, the patch panel **404** and the power supply **406** are both rack mounted. Optical communication lines from the patch panel **404** and power lines from the power source **406** are coupled to hybrid cables **20** routed to universal interface devices **310** that support active devices such as a picocell **408**, a metrocell **410**, a femtocell **412** and an ONT **414**. The ONT **414** is shown connected to the corresponding interface device **310** by a power line **415** and a separate fiber line **417**. The picocell **408**, metrocell **410** and femtocell **412** are coupled to their corresponding interface devices **310** by power lines **419** and two fiber lines **421**, **423**. In other examples, the fiber lines **421**, **423** can be replaced with twisted pair conductors for carrying electrical signals in cases where optical to electrical signal conversion occurs at the interface device **310**.

(54) Further aspects of the present disclosure relate to a remote interface device (e.g., interface device **310** that is remote from the power supply) for providing an interface between a hybrid cable and a remote active device, the interface device including a closure that houses circuitry for providing electrical power management and including circuit

protection electronics. It will be appreciated that the circuitry within the closure is adapted for providing an effective interface between a hybrid cable (e.g., hybrid cable **20**) and a remote active device. In certain examples, the closure is designed for outdoor environmental use and includes an environmentally sealed construction. In certain examples, the electrical power management circuitry eliminates the need for line power system design. For example, the electrical power management circuitry can include a DC-to-DC converter suitable for converting power carried by one of the hybrid cables **20** to a voltage and power level compatible with an active device intended to be powered with power from the hybrid cable **20**. In certain examples, the DC-to-DC converter can increase the voltage of the power carried by the hybrid cable **20** to a level compatible with the active device powered by the hybrid cable. In certain examples, the increase in voltage provided by the DC-to-DC converter compensates for voltage loss that may occur over the length of the hybrid cable. In certain examples, the DC-to-DC converter raises the voltage level to 12 volts, 24 volts or 48 volts. In certain other examples, the DC-to-DC converter decreases the voltage level to a level compatible with the active device intended to be powered by the hybrid cable. In certain examples, the power is converted so as to become compatible with a 25 watt device, a 30 watt device, or a 45 watt device. In certain examples, the closure also houses an optical-to-electrical converter that converts optical signals from the hybrid cable to electrical signals that are transmitted to the active device. In certain examples, the electrical signals and the power can be transmitted from the interface device to the active device by a twisted pair Ethernet cable so as to provide power-over-Ethernet or power-over-Ethernet-plus connectivity.

(55) As indicated above, the closure of the interface device can enclose circuit protection electronics. For example, the circuit protection electronics can include primary electrical protection that may include a gas discharge tube rated to at least 40 kAmp surge/overvoltage protection. Such structure can provide protection with respect to lightning strikes and line cross-overs. The electrical protection can also include secondary electrical protection that may be rated to 4.5 kAmp and that may include metal oxide varistor components that couple to ground in response to voltage surges. The electrical protection may also include tertiary protection that prevents voltage from rising above a predetermined level (e.g., 80 volts, or 100 volts). In certain examples, the tertiary protection can include a transient voltage suppression diode. In certain examples, fast acting fuses can be used.

(56) Cables in accordance with the principles of the present disclosure can provide power over relatively long distances. For example, using 12 gauge conductors in the hybrid cable and using conversion circuitry in the interface device that converts the optical signals and power to a PoE format, the system can provide 10 Watts of power over a length of 3,000 meters, 15 Watts of power over 2,400 meters, 20 Watts of power over 1,900 meters and 25 Watts of power over 1,600 meters. If power is provided in a non-power over Ethernet format (e.g., via an M8 plug or other power lines separate from the communications lines), 30 watt power can be provided up to 1,600 meters and 45 watt power can be provided up to 1,000 meters. A system that utilizes 16 gauge conductors and outputs power in a power over Ethernet format can provide 10 watts of power at 1,200 meters, 15 watts of power at 960 meters, 20 watts of power at 760 meters, and 25 watts of power at 640 meters. By not using a power over Ethernet format and instead keeping the power separate from the communications via a separate power line, the 16 gauge wire can provide 30 watts of power at 640 meters and 45 watts of power at 400 meters.

(57) Aspects of the present disclosure relate to interface closures that can be readily customized to meet customer requirements. In certain examples, the closures can be environmentally sealed and can include clamps for clamping hybrid cables such as the hybrid cable **20**. The closures can also include power management circuitry such as power converters (e.g., DC-to-DC power converters). The power converters can be customized to comply with the power requirements of the remote device intended to be powered by the customer. In certain examples, the power conversion circuitry can be modular and modules providing different levels of conversion can be selected and plugged into the circuit board of the closure to satisfy the customer requirement. For example, power converters capable of outputting 12, 24 or 48 volts can be used. It will be appreciated that the format of the power output from the interface closure can also be customized to meet customer needs. For example, the interface closure can be configured to output power and communications over a variety of formats such as: (a) power-over-Ethernet; (b) power-over-Ethernet-Plus; (c) separate power (e.g., via a cable terminated with an M8 plug or other configuration) and Ethernet lines (e.g., terminated with RJ45 connectors or other connectors); (d) separate fiber lines for communications and power lines for power (e.g., terminated with M8 connectors or other power connectors); (e) a hybrid cable having optical fibers for optical signals and electrical conductors for power that can be terminated with a hybrid connector or can have separate fiber and power pigtailed; or (f) a cable having twisted pair conductors for carrying communication signals and separate electrical conductors for power that can be terminated by separate RJ-style connectors for communication signals and an M8 plug for power or other connector arrangements. In the case where separate fiber lines are used, the fiber lines can be terminated with different styles of fiber optic connectors such as LC connectors, SC connectors, or other fiber optic connectors. In certain examples, the fiber optic connectors can be ruggedized and can include environmental sealing as well as twists-to-lock fastening elements such as threaded fasteners or bayonet-style fasteners. In the case of Ethernet cable, standard RJ-45 connectors or ruggedized RJ-45 connectors can be used. For pigtailed carrying only power, stranded or solid conductors can be used. Additionally, the power pigtailed can be terminated with power connectors such as M8 connectors.

(58) FIG. **16** shows a cable system **100** that can be used to transmit power and communications from a first location **102**

to an active device **104** at a second location **106**. The second location **106** is remote from the first location **102**. In certain example, the first location **102** can be a base location and the active device **104** can include wireless coverage area defining equipment. Examples of wireless coverage area defining equipment and locations where such equipment may be installed are described above. Examples of other types of active devices include cameras such as high definition video cameras.

(59) The first location **102** receives optical signals from a remote location **108** via a fiber optic trunk cable **110**. Optical fibers of the trunk cable **110** can be separated at a fan-out device **111** at the first location. Alternatively, optical power splitters or wavelength division multi-plexers can be used to split optical communications signals from the trunk cable **110** to multiple optical fibers. The fibers can be routed to a patch panel **112** having fiber optic adapters **114** (i.e., structures for optically and mechanically interconnecting two fiber optic connectors **115**). The first location **102** can also include a combined power/communication panel **116** having fiber optic adapters **117** paired with power adapters **118** (i.e., ports). Connectorized fiber optic patch cords **120** can be routed from the fiber optic adapters **114** to the fiber optic adapters **117**.

(60) The first location **102** can receive electrical power from a main power line **122**. In one example the main power line **122** can be part of a mains power system that provides 100-240 nominal volt alternating current (example frequencies include 50 and 60 Hertz). The first location **102** can include a converter **124** for converting the electrical power from the first voltage (e.g., 100v, 120v, 220v, 230v, 240v etc. nominal voltage) to a second voltage that is less than the first voltage. In one example, the second voltage is less than or equal to 60 volts and less than or equal to 100 Watts such that the output voltage complies with NEC Class II requirements. In one example, the converter **124** is an AC/DC converter that converts the electrical power from alternating current to direct current. Connectorized power cords **126** can be used to route electrical power having the second voltage from the converter **124** to the power adapters **118**. In certain examples, the combined power/communications panel **116** can include at least 18, 24, 30 or 32 fiber optic adapters paired with corresponding power adapters **118**. In certain examples, the converter **124** is large enough to provide NEC Class II compliant power through separate hybrid cables to at least 18, 24, 30 or 32 active devices. Of course, converter having smaller capacities could be used as well. Additionally, the converter **124** can be part of a voltage conversion package including overvoltage protection that provides protection/grounding in the event of lightning strikes and main crosses.

(61) A hybrid cable **20** can be used to transmit electrical power and optical communication signals between the first and second locations **102**, **106**. The hybrid cable **20** can include an outer jacket **150** containing at least one optical fiber **152** for carrying the optical communication signals and electrical conductors **154** (e.g., wires such as ground and power wires) for transmitting the electrical power having the second voltage. The hybrid cable **20** can include a first end **156** and a second end **158**. The first end **156** can include a first interface for connecting the hybrid cable to electrical power and fiber optic communication at the first location **102**. In one example, the first interface can include a power connector **160** (e.g., a plug) that connects the electrical conductors **154** to one of the connectorized power cords **126** at the power/communications panel **116**. The power connector **160** can be plugged into the adapter **118** and can be provided at a free end of a cord that extends outwardly from the outer jacket **150** at the first end of the hybrid cable **20**. The cord can contain the electrical conductors **154**. The first interface can also include a fiber optic connector **162** (e.g., an SC connector, LC connector, ST-style connector or other type of connector) that connects the optical fiber **152** to one of the patch cords **120**. The fiber optic connector **162** can plug into one of the fiber optic adapters **117** and can be mounted at the free end of a cord that contains the optical fiber **152** and extends outwardly from the outer jacket **150** at the first end of the hybrid cable **20**.

(62) The second end **158** of the hybrid cable **20** can include a second interface for connecting the hybrid cable **20** to the active device **104** such that electrical power is provided to the active device **104** and such that fiber optic communication signals can be transmitted between the first and second locations **102**, **106**. The second interface includes an interface structure **164** including a power connection location **166** and a communication connection location **168**. In one example, the interface structure **164** includes a power converter **170** for converting electrical power carried by the hybrid cable **20** to a direct current third voltage that is less than the second voltage. In one example, the third voltage corresponds to an electrical voltage requirement of the active device **104**. In one example, the power converter **170** is a DC/DC converter. In one example, the third voltage is 12V, 24V or 48V. In examples where AC current is transmitted by the hybrid cable **20**, the power converter **124** can be an AC/AC converter and the power converter **170** can be an AC/DC converter. In certain examples, the interface structure **164** can include an optical-to-electrical converter for converting the communications signals carried by the optical fiber **152** from an optical form to an electrical form. In other examples, optical-to-electrical conversion can be performed by the active device **104** or can take place between the active device **104** and the interface structure **164**.

(63) In one example, the interface structure **164** includes a converter interface that allows power converters **170** with different conversion ratios to interface and be compatible with the interface structure **164**. The conversion ratio of the particular power converter **170** used can be selected based on factors such as the voltage requirement of the active device **104** and the length of the hybrid cable **20**. The power converters **170** can have a modular configuration can be installed within the interface structure **168** in the field or in the factory. In one example, the power converters **170** can have a “plug-and-play” interface with the interface structure. The modular configuration also allows the power

converter **170** to be easily replaced with another power converter **170**, if necessary. In certain examples, the interface structure **164** can include overvoltage protection and grounding arrangements such as fuses, metal oxide varistors, gas tubes or combinations thereof.

(64) In one example, the electrical power having the third voltage can be output to the active device **104** through the power connection location **166**. The power connection location **166** can include a power connector, a power port, a power cord or like structures for facilitating connecting power to the active device **104**. In one example, the power connection location **166** can have a modular configuration that allows interface connectors having different form factors to be used.

(65) In one example, the communications signals can be transferred between the hybrid cable **20** and the active device through the communication connection location **168**. The communication connection location **168** can include a connector, a port, a cord or like structures for facilitating connecting to the active device **104**. In one example, the communication connection location **168** can have a modular configuration that allows interface connectors having different form factors to be used. In the case where the optical to electrical converter is provided within the interface structure **164**, the connection location can include electrical communication type connectors (e.g., plugs or jacks) such as RJ style connectors. In the case where the optical to electrical converter is provided at the active device **104**, the communication connection location **168** can include fiber optic connectors and or fiber optic adapters (e.g., SC connectors/adapters; LC connectors/adapters, etc.). In certain examples, ruggedized, environmentally sealed connectors/adapters can be used (e.g., see U.S. Pat. Nos. 8,556,520; 7,264,402; 7,090,407; and 7,744,286 which are hereby incorporated by reference in their entireties. It will be appreciated that when the active devices include wireless transceivers, the active devices can receive wireless signals from the coverage area and such signals can be carried from the active devices to the base station **11** via the hybrid cables. Also, the active devices can covert signals received from the hybrid cables into wireless signals that are broadcasted/transmitted over the coverage area.

(66) In one example, the second voltage is less than the first voltage and greater than the third voltage. The third voltage is the voltage required by the active device at the second location. In one example, the second voltage is sufficiently larger than the third voltage to account for inherent voltage losses that occur along the length of the hybrid cable.

(67) Various modifications and alterations of this disclosure may become apparent to those skilled in the art without departing from the scope and spirit of this disclosure, and it should be understood that the scope of this disclosure is not to be unduly limited to the illustrative examples set forth herein.

Claims

1. A method for providing electrical power and communication signals to an outdoor located active device, the method comprising: converting the electrical power from an AC voltage to a DC voltage at a first location; transmitting the electrical power having the DC voltage through a hybrid cable from the first location to an interface device at a second location, the interface device being adapted for outside environmental use and having an environmentally sealed construction; transmitting the communication signals in an optical form through the hybrid cable from the first location to the interface device at the second location; converting the electrical power at the interface device to an active device compatible DC voltage; converting the communication signals at the interface device from the optical form to an electrical form; and transmitting the electrical power having the active device compatible DC voltage and the communication signals in the electrical form from the conversion device to the outdoor located active device via an Ethernet cable having twisted pair conductors.
2. The method of claim 1, wherein the outdoor located active device comprises at least one of a wireless transmitter device and a camera device.
3. The method of claim 1, wherein the conversion device includes a housing having a cover that can be opened and closed, wherein an interior of the housing is accessible by opening the cover, and wherein the cover is required to be opened to connect the Ethernet cable to optical-to-electrical conversion circuitry and power management circuitry within the interior of the housing.
4. The method of claim 1, wherein the hybrid cable is routed outdoors.
5. The method of claim 1, wherein the hybrid cable is plenum rated.
6. A powered fiber system comprising: a hybrid cable having an electrical conductor and an optical fiber; a first interface device including voltage conversion circuitry for converting electrical power from a AC voltage to a first DC voltage, the first interface device having a power interface for sending electrical power having the first DC voltage through the electrical conductor of the hybrid cable and an optical interface for sending optical signals through the optical fiber of the hybrid cable; an Ethernet cable including first and second wires arranged in a twisted pair configuration; a second interface device that is adapted for outside environmental use and that has an environmentally sealed construction, the second interface device including a housing defining an interior, the housing including a first port for allowing the hybrid cable to be routed into the interior of the housing and a second port for allowing the Ethernet cable to be routed into the interior of the housing, the second interface device including electrical power management circuitry positioned within the interior of the housing, the electrical power management circuitry being configured to be coupled between the electrical conductor of the hybrid cable and the twisted pair wires of the Ethernet

cable, the electrical power management circuitry having a DC-to-DC voltage converter for converting the electrical power received from the hybrid cable to a voltage suitable for transmission over the first and second wires of the Ethernet cable, the second interface device also including optical-to-electrical conversion circuitry positioned within the interior of the housing for connecting the optical fiber of the hybrid cable to the first and second wires of the Ethernet cable, the optical-to-electrical conversion circuitry being configured to convert the optical signals from the optical fiber into electrical signals that are transmitted over the first and second wires of the Ethernet cable.

7. The powered fiber system of claim 6, wherein the housing has a cover that can be opened and closed, wherein the interior of the housing is accessible by opening the cover, and wherein the cover is required to be opened to connect the Ethernet cable to the optical-to-electrical conversion circuitry and the electrical power management circuitry within the interior of the housing.

8. The powered fiber system of claim 6, wherein the hybrid cable is plenum rated.

9. The powered fiber optic system of claim 6, further comprising circuit protection circuitry in the interior of the housing for providing current surge protection.

10. The powered fiber optic system of claim 9, wherein the circuit protection circuitry provides primary, secondary and tertiary protection.

11. The powered fiber optic system of claim 9, wherein the circuit protection circuitry includes a transient voltage suppression diode.

12. The powered fiber optic system of claim 6, further comprising cable clamps configured to fix the hybrid cable and the Ethernet cable to the housing respectively at the first and second cable ports.
